FROM THE

EDITOR-IN-CHIEF

Spying has probably been going on from time immemorial. Countries spying on each other, governments spying on their citizens, political parties spying on each other and companies spying for industrial secrets. However, with the advent of technology, especially the internet, it is less dependent on human intelligence and more on technical intelligence. Now that we are all so interconnected through our various devices, we are vulnerable to the invasion of our private spaces. Even innocent-looking devices like Alexa know more about you than you ever know. However, the latest public reckoning of spying, which has stirred the government into vigilance, has come from elsewhere.

Late in January, a Chinese 'weather balloon' spotted high over American skies—borne along by high winds all the way from Alaska to South Carolina, at an incredible 60,000 feet—provoked US President Joe Biden to shoot it down. But the object presently spooking India's strategic experts is much closer at hand: the innocuous-looking closed-circuit television camera, or CCTV, the sort that peers at you in an elevator or an ATM kiosk.

Most CCTVs in the world are of Chinese origin. The leading global suppliers are Hikvision and Dahua, two firms partly owned by the Chinese government. Concerns that these could be sneaky espionage tools had prompted an alarmed Washington to push back against Beijing's creeping spy games. The US banned Hikvision, Dahua and Huawei equipment from use in government and critical infrastructure installations in 2019. Last year, the UK followed suit; Australia joined the list last fortnight. The Chinese whispers reached India too after the clash in the icy heights, and we have woken to the fact that CCTV cameras of Chinese make dominate our market—accounting for up to 80 per cent of domestic applications and over *98 per cent* of government installations, including military compounds.



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India's strategic establishment, long trained to think in terrestrial terms, may have been slow to wake up to the way the war has been scaled up to a more rarefied level. The threat is not limited to the icy heights of Galwan or Yangtse. The intrusion may be less obvious than boots on the ground, and more abstract than 38,000 sq. km of territory. But the Chinese eye on us is omnipresent, rendered even more inscrutable by the miniaturising magic of the digital age. If the alarmist version of the story is taken at face value, it's not just one eye but all of two million eyes, watching us everywhere, from mundane street corners to high-value defence installations.

About a million stand unblinkingly in government institutions, another million cover India's civic life. Count everything from private bungalows to the 150,000 CCTV cameras on Delhi's streets, to those used on the Delhi Metro or by the SPG, right up to highly sensitive defence sites via Bharat Electronics Limited and DRDO clientship. And as Senior Editor Pradip R. Sagar's cover story this week reveals, an internal note of the Ministry of Defence (MoD) in May 2021 noted grimly that surveillance cameras of Chinese make were installed in all Indian naval establishments.

CCTVs seem to have escaped global strategic attention earlier because of their purportedly innocent nature. The very term 'closed-circuit television camera' conjures up a simplistic vision: an all-seeing eye that never sleeps but the data generated by it flowing only within a tight, restricted loop. What if the circuit were not as closed as everyone imagined? Increasingly, a CCTV is more than just a camera wired to a local area network. Today's Video Surveillance Systems often rely on centrally controlled Network Video Recorders (NVR) for storage, and these work with Internet Protocol cameras, not the old analog ones. Besides offering technically advanced features like higher definition, varifocal or pan-tilt-zoom imaging and onboard video analytics, IP cameras are—as the name suggests—*networked* devices. They are made to receive control data and send visual data. Specifically, they come embedded with a sophisticated integrated circuit called a System-on-Chip (SoC). Again, as the name tells you, this is a whole system—a powerful assembly of functions like the CPU, graphics processor, memory and, more crucially, WiFi or cellular radio modems—all crunched into a chip that can be as small as a shirt button. Modem connectivity means its capabilities can be switched on remotely.

A curious name in the global SoC market is HiSilicon, wholly owned by Huawei, the company that gained notoriety amidst allegations that it served as a cat's paw for Chinese espionage. This led to it being banned from several 5G markets, including India. Established only in 2004, heavy R&D investment propelled HiSilicon to the top of the global game—it found its way even into US-made Honeywell cameras—before a 2020 export control order by the US saw it debarred from access to American chip design software and Taiwan-based "foundries". But a silently ticking HiSilicon chip is in our midst everywhere. Experts had been warning of the spectre of Chinese cyber-snooping in all its forms for well over a decade. But this micro-device has only just been crowned as a likely protagonist in the belated bout of alarm that presently has New Delhi in its grip, with genuine fears about the suspected flow of sensitive military data to Beijing. "You don't need to send spies across the border anymore—these CCTVs become the eyes of any country wanting to do mischief," Lt Gen. Rajesh Pant (retd), the National Cyber Security Coordinator for India's National Security Council, told our team doing the cover story. "They are the best tools for techint [technical intelligence]. The unchecked proliferation of such devices across our country is a serious security risk."

Of course, two million CCTVs cannot be suddenly thrown into the Arabian Sea without throwing our systems totally awry, so what can be a solution? A three-pronged way lies ahead of India. To begin with, a scaling back has begun. The Southern Naval Command is phasing out Hikvision CCTVs. We also need legal barriers against digital spies sneaking in through the backdoor via 'Made in India' partnerships. Simultaneously, cyber experts are combating technology on its own turf, firstly by reprogramming Chinese chips with Indian software. Till such time as India can manufacture its own high-quality CCTVs, testing systems must be installed that can trace the presence of sneaky micro-devices in any foreign equipment, whether of Chinese make or otherwise. Espionage is still all cloak-and-dagger drama, but more with chips and net.

(Aroon Purie)